
ECON 212 — Principles of Microeconomics

Handout: Consumer and Producer Surplus

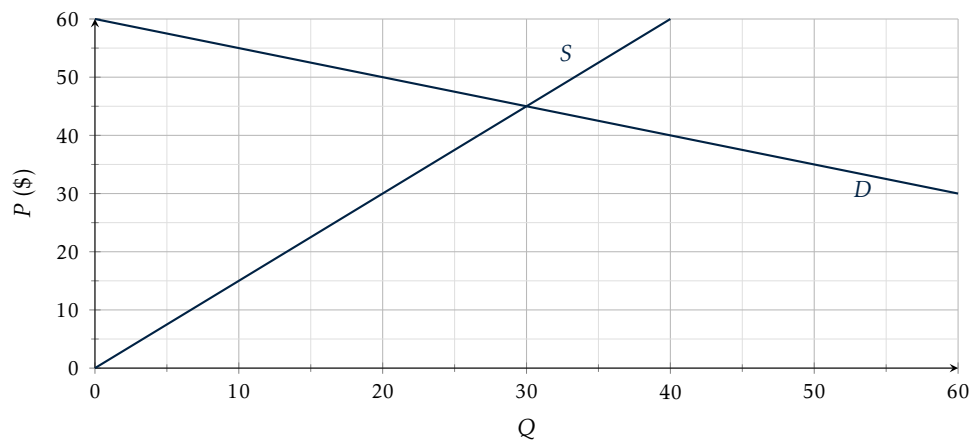
Fall 2026

Work with the person next to you; shade directly on the figures. Consumer-side regions are shaded orange and labelled *CS*; producer-side regions are shaded blue and labelled *PS*. In Part II a solid fill is the intensive margin and a hatched fill is the extensive margin.

Part I. Surplus at the Equilibrium

Problem 1. Driveway snow removal

Inverse demand: $P = 60 - \frac{1}{2}Q$ Inverse supply: $P = \frac{3}{2}Q$



(a) Find q^* and p^* .

(b) Find the highest price any buyer will pay and the lowest price any seller will accept.

Highest price any buyer will pay _____

Lowest price any seller will accept _____

(c) Shade CS on the figure and compute it.

(d) Shade PS on the figure and compute it.

(e) Compute TS .

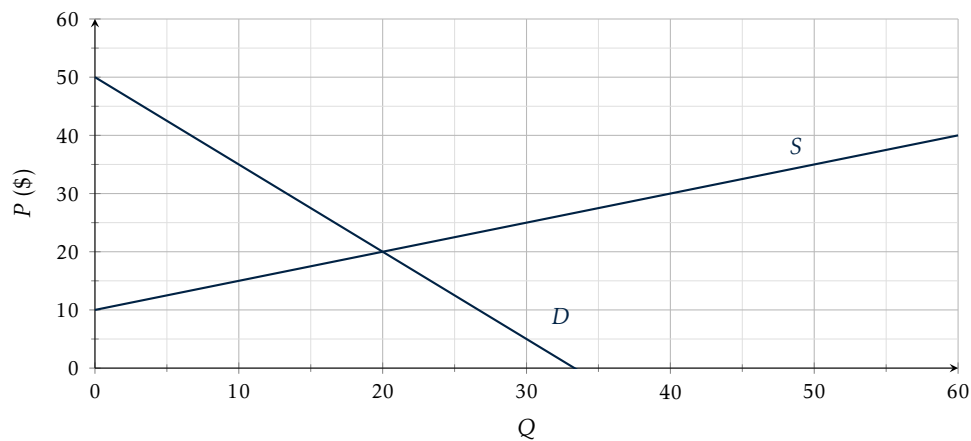
CS _____

PS _____

$TS = CS + PS$ _____

Problem 2. Bottled water at the stadium

Inverse demand: $P = 50 - \frac{3}{2}Q$ Inverse supply: $P = 10 + \frac{1}{2}Q$



(a) Find q^* and p^* .

(b) Find the highest price any buyer will pay and the lowest price any seller will accept.

Highest price any buyer will pay _____

Lowest price any seller will accept _____

(c) Shade CS on the figure and compute it.

(d) Shade PS on the figure and compute it.

(e) Compute TS .

CS _____

PS _____

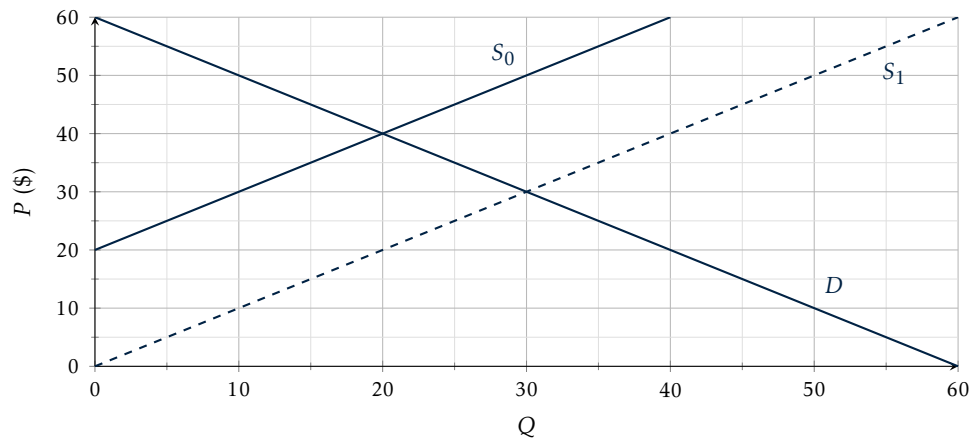
$TS = CS + PS$ _____

Part II. Intensive and Extensive Margins

Problem 3. Coffee beans, from the buyer's side

Inverse demand: $P = 60 - Q$ Inverse supply: $P = 20 + Q$ Start: $q_0 = 20, p_0 = \$40$

A new roasting machine cuts every seller's cost by \$20, so inverse supply shifts down to $P = Q$.



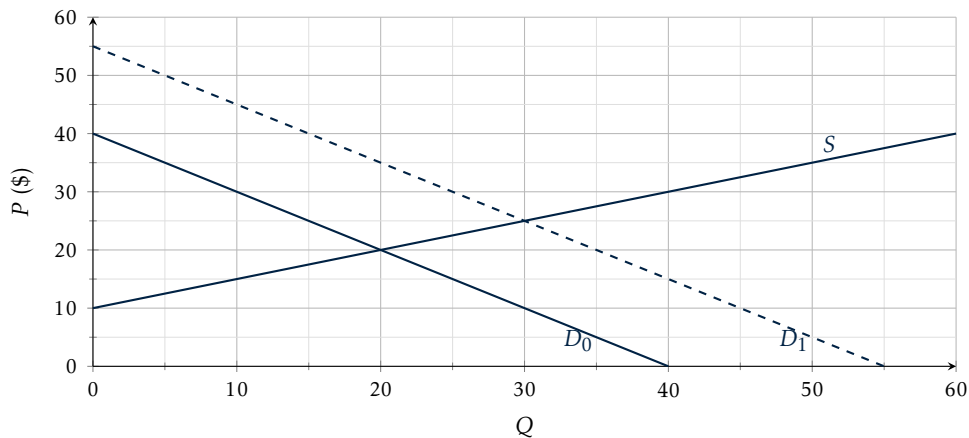
- (a) Find the new equilibrium q_1 and p_1 .
- (b) Compute the gain on the intensive margin: the surplus change on the q_0 units that were already trading.
- (c) Compute the gain on the extensive margin: the surplus earned on the $q_1 - q_0$ new units.
- (d) Compute the total change in CS, and confirm it equals (b) + (c).

Intensive margin	_____
Extensive margin	_____
Total change in CS	_____

Problem 4. Donuts, from the seller's side

Inverse demand: $P = 40 - Q$ Inverse supply: $P = 10 + \frac{1}{2}Q$ Start: $q_0 = 20, p_0 = \$20$

A new dorm opens across the street, so inverse demand shifts up by \$15, to $P = 55 - Q$.



- (a) Find the new equilibrium q_1 and p_1 .
- (b) Compute the gain on the intensive margin: the surplus change on the q_0 units that were already trading.
- (c) Compute the gain on the extensive margin: the surplus earned on the $q_1 - q_0$ new units.
- (d) Compute the total change in PS , and confirm it equals (b) + (c).

Intensive margin	_____
Extensive margin	_____
Total change in PS	_____

Part III. Surplus When $p \neq p^*$

Assume efficient rationing throughout: the units that trade go to the highest-value buyers and come from the lowest-cost sellers.

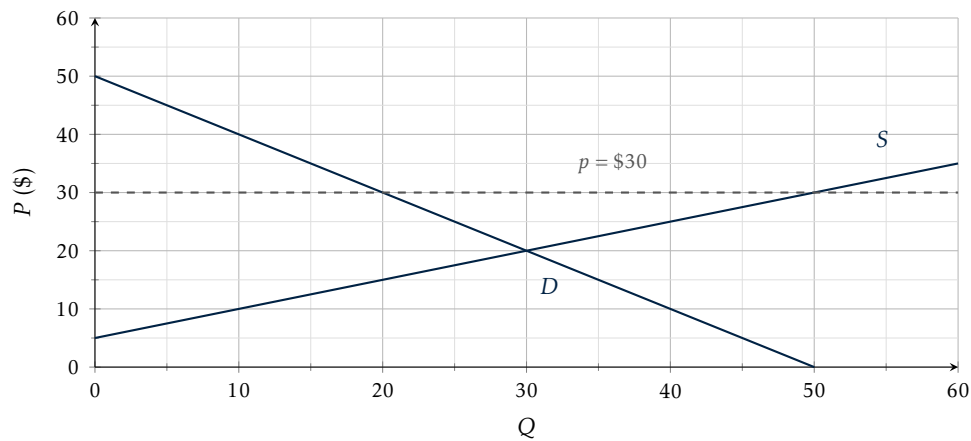
Problems 5 and 6 use the same market:

$$\text{inverse demand: } P = 50 - Q, \quad \text{inverse supply: } P = 5 + \frac{1}{2}Q.$$

Take as given that $q^* = 30$ and $p^* = \$20$, and that at that equilibrium $CS = \$450$, $PS = \$225$, and $TS = \$675$.

Problem 5. A price above p^*

The price in this market is held at \$30.



- (a) At the given price, compute Q_D and Q_S . How many units actually trade?

Q_D at $p = \$30$ _____

Q_S at $p = \$30$ _____

Units actually traded: _____

- (b) Shade CS on the figure and compute it.

- (c) Shade PS on the figure and compute it.

- (d) Compute TS, and compute how much TS falls short of its value at p^* .

CS _____

PS _____

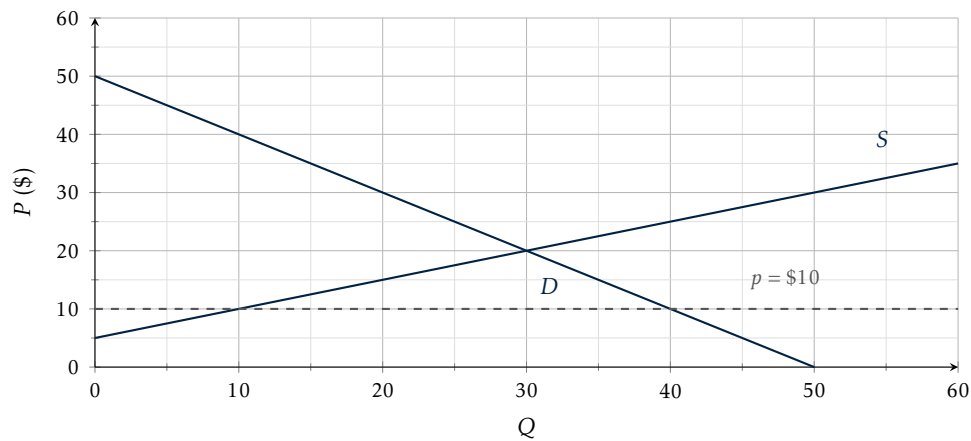
TS _____

TS at p^* _____

TS falls short of its value at p^* by _____

Problem 6. A price below p^*

The price in this market is held at \$10.



- (a) At the given price, compute Q_D and Q_S . How many units actually trade?

Q_D at $p = \$10$ _____

Q_S at $p = \$10$ _____

Units actually traded: _____

- (b) Shade CS on the figure and compute it.

- (c) Shade PS on the figure and compute it.

- (d) Compute TS , and compute how much TS falls short of its value at p^* .

CS _____

PS _____

TS _____

TS at p^* _____

TS falls short of its value at p^* by _____